

Session Objectives

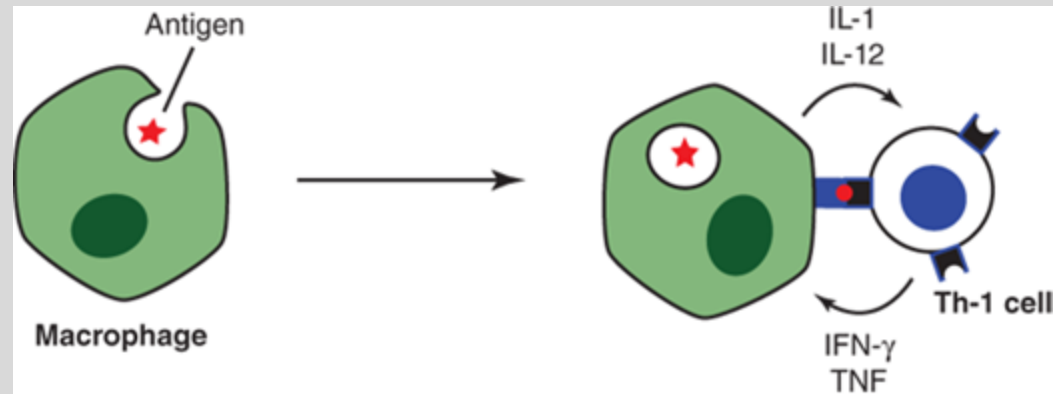
- Discuss the four hypersensitivity reactions, with the main emphasis on Type 1 hypersensitivity reactions.
- Identify the clinical manifestations of allergies & the differential diagnosis of allergic reactions
- Recognize when a patient is having an anaphylactic reaction
- Treatment of anaphylaxis and non-anaphylactic allergic reactions

Type IV Hypersensitivity–Cell Mediated

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- Previously sensitized T1 helper
- Cells recognize an antigen (Ag Presenting cell) and release cytokines to recruit more T1 helper cells and mononuclear cells
- Inflammatory reaction
- Ex: Contact Dermatitis



Source: W. Levinson, P. Chin-Hong, E.A. Joyce, J. Nussbaum, B. Schwartz:
Review of Medical Microbiology & Immunology: A Guide to Clinical Infectious
Diseases, Seventeenth Edition: Copyright © McGraw Hill. All rights reserved.

Late Phase

- About 6 hours after exposure
- Due to mediators (leukotrienes) synthesized after cell degranulates
- Influx of inflammatory cells (neutrophils and eosinophils)
- Biphasic Reaction – reaction meeting anaphylactic criteria within 48 hours of previous reaction without re-exposure to Ag

Symptoms

- Skin – Flushing, Pruritis, Urticaria (hives)
- Respiratory – Rhinitis, Stridor, Wheezing
- GI – Nausea, Vomiting, Diarrhea
- ENT – Uvular edema, Tongue Swelling, Oral pharyngeal edema
- Neuro – Headache, Seizure
- CV – Chest Pain, Hypotension

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Source: Kevin J. Knoop, Lawrence B. Stack, Alan B. Storrow,
R. Jason Thurman: The Atlas of Emergency Medicine, 5e
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Table 6. Differential Diagnosis for Allergic Reactions

- **Life-Threatening Causes**

- Angioedema
- Acute coronary syndromes
- Arrhythmia
- Airway obstruction or foreign body
- Asthma and other atopic diseases
- Sepsis and septic shock
- Pneumothorax
- Scombroid poisoning
-

- **Non-Life-Threatening Causes**

- Anxiety
- Panic attack
- Vasovagal reaction/syncope
- Dermatitis
- Gastroenteritis
- Chronic urticaria
- Mastocytosis or other mast cell disorders
- Liver dysfunction

Zeke A. Sudhir A. Management of Allergic Reactions and Anaphylaxis in the Emergency Department. *Emerg. Med Practice* 2022, 24 (7): 1-24.

Diagnosis of Anaphylaxis

- 1. Acute onset of mucocutaneous Symptoms (flushing, pruritis, urticaria, angioedema) plus one of the following
 - Wheezing, stridor, hypoxia
 - Hypotension or end-organ damage
- 2. Two or more of the following that occur rapidly after exposure to known or suspected allergen
 - Mucocutaneous Involvement
 - Respiratory Symptoms
 - Hypotension or end-organ hypoperfusion
 - GI Symptoms (pain, nausea, vomiting)

Diagnosis of Anaphylaxis

- 3. Hypotension within minutes to hours after exposure to a known allergen
 - Hypotension = Anaphylactic Shock

Treatment

- Airway Protection
- **Epinephrine** —
 - 0.3-0.5 mg IM (in the thigh)
 - Peds: 0.01 mg/kg IM
 - Only evidence proven medication for anaphylaxis
 - Increases Peripheral Vascular Resistance and Cardiac Output
 - Stabilizes mast cells
 - Reverses Bronchoconstriction and mucosal edema

Treatment

- Refractory to IM Epi – use IV
- Decontamination – Remove triggers
 - Ex: insect stinger
- Albuterol if wheezing only – alleviate bronchospasm
- IV crystalloid infusion if hypotensive
- Corticosteroids (evidence low) – anti-inflammatory and may help prevent biphasic reactions

Treatment

- The Anti-histamines:
 - No evidence to support life-threatening aspects of anaphylaxis
 - Helps w/ cutaneous symptoms
- Is your pt on a Beta-blocker:
 - May need Glucagon

Disposition

- Low, risk pt, Resolved Anaphylaxis
 - Observe and discharge if remain stable
 - Rx for epinephrine auto-injector
 - Consider steroids or anti-histamine
 - Rx for symptoms relief
 - Follow up w/ primary care physician or allergist

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Disposition

- High Risk, Mild to Moderate Anaphylaxis
- Persistent symptoms, increased risk of death from anaphylaxis (Congestive Heart Failure, Chronic Kidney Disease, asthma, hx of severe allergic reactions, pts on Beta Blockers)
- Prolonged ED Observation or Admit

Non-anaphylactic Allergic Reactions

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- 90% of pts have urticaria
 - Ask about potential exposures
 - New meds? Soaps? Foods? Exposures? Family hx of allergic reactions?
- Not all rashes – allergic reactions
 - Ex: virus in children



Source: Joseph Loscalzo, Anthony Fauci, Dennis Kasper, Stephen Hauser, Dan Longo, J. Larry Jameson: Harrison's Principles of Internal Medicine, 21e
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Non-anaphylactic allergic reactions

- Tx:
 - Antihistamines – symptoms
 - H1/H2 Blockers (cutaneous symptoms)
 - Monitor for progression to anaphylaxis
 - + / - Corticosteroids
 - Decrease itch
 - Risk versus Benefit in mild/moderate reactions
 - Severe – Give Steroids

Summary Slide

- Allergies/Anaphylactic Reactions are Type 1 Hypersensitivity Reactions
- Epinephrine is the life-saving medication for anaphylaxis
- Avoid re-exposure to known allergens when possible